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Assignment #1 | Mathematics - III (MTH 231) | Date: August 12, 2024
Branch - Electrical Engineering | Section - I and II

1. When working with numerical algorithms, what are the different ways to measure error of approximation? Explain with examples. Which of these methods give a more accurate understanding of error and why?

2. Regula Falsi method as well as Secant method have the same iteration formula for guessing the next root. What is the difference between them? Explain with reasons.

3. I want to use Newton Raphson method to find the root of a function $f(x)$. What kind of functions does $f(x)$ have to be so I can use the method?

4. Differentiate between bracketing methods (like Bisection and Regula Falsi) and iteration methods (like Newton-Raphson and Secant). Discuss in detail the advantages and disadvantages of each class of methods.

5. Someone asked me to find the double root of a degree 7 polynomial $p(x)$. So I assumed an initial value x_0 and applied Generalised Newton-Raphson method as follows

$$\begin{aligned} x_1 &= x_0 - 2\frac{f(x_0)}{f'(x_0)} \\ x_2 &= x_1 - 2\frac{f(x_1)}{f'(x_1)} \\ x_3 &= x_2 - 2\frac{f(x_2)}{f'(x_2)} \\ &\vdots \qquad\qquad \vdots \end{aligned}$$

   On analysing the resulting sequence, I found that the approximate root is 2.3451. I hence, concluded that this is the required double root. Am I correct? If yes, why? If no, why not?

6. State whether the following statements are True or False. Explain your reasons in detail for each.
   (a) My friend calculated the roots of a polynomial as 0.5631 and 0.7245 correct to four decimal places. He claims he used Graeffe's root squaring method.
   (b) I can use Graeffe's root squaring method to estimate that the roots of $x^2 - 361$ are $+19$ and -19.

7. Given a polynomial $p(x)$ has roots r_1, r_2, r_3. Construct a polynomial that has roots r_1^4, r_2^4, r_3^4. Use this to find a polynomial with roots 2^4, 7^4 and $(-3)^4$.

8. Find the value of $\tan\frac{\pi}{12}$ correct upto 4 decimal places using
   (a) Regula Falsi Method
   (b) Newton-Raphson

   Hint : *Finding value of $f(x)$ at point x_0, is equivalent to finding the root of $g(x) = x - f(x_0)$.*

9. Find a real root of the equation $e^x - x - 3$ correct upto four decimal places using
   (a) Bisection Method
   (b) Regula Falsi method

   Which method is better? Why?

10. Given $x^3 - 4|x| + 2$ has 3 real roots. Find all three of them, correct upto four decimal places.

11. Evaluate the solution of $\ln x - x + 4 = 0$ correct to four decimal places using the Secant method.

12. Approximate the value of $(\ln 5)^{\frac{1}{6}}$ correct upto four decimal places.

13. Solve the equation $x - \cos x = 2$ for x correct upto four decimal places.

14. Verify $\pi = 3.1417$ using Newton-Raphson method.

15. Find a root of $|\csc x| - |\tan x|$ in the first quadrant correct to four decimal places.

16. Find a root of $|\cos^3 x| - |\sin^5 x|$ correct to four decimal places.

17. Given the polynomial $200x^3 - 395x^2 + 248x - 48$ has a single root and a double root - both in $(0, 1)$. Find the double root correct to four decimal places.

18. Given the polynomial $2187x^5 + 1944x^4 - 1269x^3 - 1242x^2 + 180x + 200$ has a triple root and a double root in the interval $(-1, 1)$. Find both the roots correct to four decimal places and identify which is the triple root and which is the double root.

19. Find all the roots of the polynomial $125x^3 - 1385x + 1596$ correct to four decimal places.

20. Given the polynomial $x^3 + 5x^2 - 29x - 105$. Find all the roots correct upto four decimal places.